ROADSAATHI · TOWING & ROADSIDE ASSISTANCE PLATFORM

Business & Project Plan

Market, competitors, laws, risks, AI roadmap, financial model, go-to-market and execution plan

Audience	For management review
Version	1.0 · 4 November 2026
Source	Generated from the Markdown in `docs/` of the TowingService repository

> Confidential. Third-party figures are cited; RoadSaathi projections are planning assumptions. Legal and tax points must be confirmed by qualified advisers.

Contents

01 RoadSaathi — Business & Project Plan

02 Chapter 2 — Competitor analysis (India and global)

03 Chapter 3 — Legal and regulatory map

04 Chapter 4 — Biggest obstacles and risk register

05 Chapter 5 — AI features roadmap

06 Chapter 6 — Financial model and unit economics

07 Chapter 7 — Go-to-market plan and KPIs

08 Chapter 8 — Execution roadmap and launch checklist

RoadSaathi — Business & Project Plan

Prepared for: management review · **Version:** 1.0 · **Date:** November 2026 **Product:** RoadSaathi, a towing and roadside-assistance platform for Delhi, with the name configurable from the admin panel **Status:** product built and deployed (Vercel + Render + Neon); not yet launched to paying customers

> **How to read this pack.** This chapter is the summary. Every claim in it is expanded in a later chapter: competitors (ch. 2), laws (ch. 3), risks (ch. 4), AI roadmap (ch. 5), financial model (ch. 6), go-to-market (ch. 7) and the execution roadmap (ch. 8). Figures from outside sources carry a citation. Every RoadSaathi forecast is a **planning assumption**, labelled as such, and is not a promise.

---

1. Executive summary

The problem. A breakdown in Delhi still ends with a phone search: "towing near me" on Google or Justdial, calls to strangers, a price agreed on the roadside, cash, no receipt, and no idea when the truck will arrive. Most of India's roadside assistance (RSA) is sold as a free add-on to car insurance or a new-car warranty. The assistance companies behind it subcontract the actual jobs to the same unorganised local tow operators. Supply is fragmented, pricing is opaque, and nobody owns the customer experience end to end.

What we built. RoadSaathi is a working, production-grade platform with three sides:

- **Customers** book in four steps on a mobile-first bilingual (English/Hindi) website. They see an upfront GST-inclusive fare before confirming, track the truck live, get a trip PIN so only the right driver can start the job, and receive a GST tax invoice automatically.
- **Partners** (independent tow-truck owners and fleet companies) apply online, are reviewed, and e-sign a personalised agreement. They then get jobs through a driver app with live GPS and an earnings ledger.
- **Operations** run everything from one admin back-office: dispatch board, partner approval, pricing, service areas, legal pages, payouts, invoices, analytics and audit log. No code changes are needed.

The engineering is already done: about 10,600 lines of TypeScript across 44 pages, 9 API routes, a 29-table database, a unit-tested pricing engine and booking state machine, and two live deployments.

The opportunity. India had roughly **US$1.8 bn** of vehicle roadside-assistance revenue in 2024, forecast to reach **US$2.85 bn by 2030 (≈7.9 % CAGR)**. Towing is the largest segment (Grand View Research). Delhi alone has **87.6 lakh registered vehicles** (up 7.9 % in 2025-26), adding about 8.5 lakh a year (Economic Survey of Delhi via Millennium Post).

How we win. We start in one dense city with an excellent consumer experience: upfront price, live tracking, verified partners and a GST invoice. That same supply network and software then becomes a **fulfilment layer that insurers, OEMs, dealers and assistance companies plug into**. The world's profitable roadside businesses (AA and RAC in the UK, at ~30–39 % EBITDA margins) are subscription and B2B businesses. The loss-makers are thin-margin middlemen with high customer-acquisition costs (Urgently in the US). Our plan is designed around that lesson (chapter 2).

What we need. The decisions in §9 below, and a launch budget sized in chapter 6. The base-case plan needs roughly **₹2.5–3 crore over 24 months** to reach EBITDA break-even in year 3. A self-funded "lean pilot" variant costs about ₹50 lakh for the first six months.

---

2. The problem in detail

Who	Today's experience	What it costs them
Vehicle owner	Searches online, calls 3–5 numbers, negotiates on the spot, no ETA, cash only, no invoice	Time stuck on the road (often at night or in rain), overcharging, unsafe handling, no insurance paperwork
Tow operator / fleet	Leads come from Justdial listings, police contacts and word of mouth; trucks sit idle part of the day	Low utilisation, unpredictable income, no digital records, hard to get B2B contracts
Insurers, OEMs, dealers, RSA companies	Subcontract to local operators through phone-based control rooms	Slow, untracked, unverifiable jobs, and the poor experience reflects on *their* brand

3. What has been built (product inventory)

Full detail: `docs/product/FEATURE-INVENTORY.md` (docs/product/FEATURE-INVENTORY.md).

Area	Built and working	Gaps before scale
Customer website	Home, 9 services, 6 vehicle classes, pricing, coverage and locality SEO pages, FAQ, contact and call-back, EN/HI	Native mobile app (the PWA works today)
Booking	4-step wizard, map search, route distance, instant GST-inclusive quote, scheduling, OTP sign-in, trip PIN, live tracking, cancel, review	Photo upload of vehicle condition; automated cancellation-fee charging
Payments	Cash/UPI to driver (live); Razorpay online payment built behind a feature flag	Razorpay live-mode KYC; refund workflow
Partners	Online application (individual or company), admin review, contract templates, personalised agreement, e-sign with email OTP (time and IP recorded), fleet adds drivers and trucks, payout details	KYC document upload and verification (DL, RC, insurance)
Driver app	Online/offline, live GPS, job offers with countdown, PIN-verified start, cash collection, earnings ledger	Push notifications (5-second polling today)
Dispatch	Nearest-eligible-driver rounds, atomic accept, auto-expiry, next round, manual assign, scheduled release	Traffic-aware ETA; ML acceptance scoring (chapter 5)
Money and tax	Paise-exact pricing engine, night/surge, CGST+SGST/IGST split, FY invoice numbering, SAC code, commission ledger, settlement marking	Bank payout automation (e.g. RazorpayX), TDS/TCS reporting
Back-office	Dashboard, bookings, dispatch board, partners, users and staff roles, every setting editable, leads, payouts, invoices, analytics funnel, notification log, audit log	Error monitoring (Sentry), CI pipeline
Compliance	Legal page system (EN/HI, versioned), grievance page, audit log, hashed OTPs, rate limits, security headers	Consent capture, cookie consent, account deletion and data export (DPDP), finalised legal text (chapter 3)

Why this matters: the platform's riskiest technical parts are finished, tested and deployed: the pricing maths, dispatch, state machine, invoicing and partner contracts. The remaining work is mainly **launch operations** (partners, legal, payments, support) plus a short list of engineering gaps sized in chapter 8.

4. Business model

Revenue line	How it works	When
Consumer commission (marketplace take rate)	Customer pays the published rate-card fare plus GST. The platform keeps a commission on the taxable value, 20 % by default and configurable per partner	Launch
B2B fulfilment	Insurers, RSA companies, dealers, fleet operators and car-rental firms send jobs through an API or portal at contracted rates. We dispatch from our verified network and earn the spread or a per-job fee	Month 4+
Operator SaaS	Tow operators use our dispatch, ledger and GST invoicing for their *own* customers for a monthly fee (₹999–1,999)	Month 6+
RSA membership	Annual plan (e.g. ₹999–1,499/yr) covering a set number of call-outs: recurring, predictable revenue, the AA/RAC model	Year 2
White-label	Brand name, colours, legal text and domain are configurable, so a dealer group or insurer can run "their" towing app on our platform	Year 2

Unit economics (planning assumptions, chapter 6). Blended job value is ₹1,400 before GST. A 20 % take rate earns ₹280. After payment, messaging, support and a damage reserve, the contribution is about **₹210 per job before marketing**. Paid search alone can cost ₹300–500 per *first* booking, so a pure-consumer model loses money on the first job. **Repeat use, B2B volume and memberships are what make the model work.** This is the central commercial insight of the plan.

5. How RoadSaathi is different

Dimension	Unorganised local towers	Insurance/OEM RSA (Allianz, Europ, TVS Auto Assist)	Justdial / listings	**RoadSaathi**
Price known before booking	No (negotiated)	Free if covered; otherwise opaque	No	**Yes: published rate card, GST shown**
Live tracking and ETA	No	Partial / call-centre updates	No	**Yes**
Verified, contracted partners	No	Yes (subcontracted)	No	**Yes: reviewed, e-signed, rated**
Job-start security	No	Varies	No	**Trip PIN**
GST tax invoice	Rarely	Insurer-side	No	**Automatic, emailed**
Hindi-first experience	Phone only	Call centre	Partly	**Full EN/HI site**
Works without insurance cover	Yes	**No** (members only)	Yes	**Yes**
Software for the operator	No	No	Listing only	**Driver app, fleet portal, ledger**
Can be white-labelled for B2B	No	N/A	No	**Yes (brand is a setting)**

Defensibility over time comes from network density (more trucks per km² means faster arrival), operator software that operators rely on day to day (switching cost), B2B contracts, and data. That data includes arrival times, demand heatmaps and damage evidence, which power the AI features in chapter 5.

6. Market sizing (top-down, indicative)

Layer	Basis	Estimate
TAM — India roadside assistance	Grand View Research, 2024 revenue	≈ US$1.8 bn (≈ ₹15,000 cr), 7.9 % CAGR to 2030
SAM — Delhi-NCR towing and on-spot assistance	NCR's share of India's vehicle stock and RSA spend; *planning assumption 5–7 %*	≈ ₹750–1,050 cr / yr
SOM — RoadSaathi year-3 target	≈ 3.1 lakh jobs/yr (consumer, B2B and member call-outs), ≈ ₹43 cr GMV (chapter 6)	≈ 4–5 % of SAM

The SAM share is an assumption to be checked against primary data (operator interviews, insurer claim volumes) in the first 60 days of the pilot.

7. Team and organisation (first 12 months)

Role	Count	Why
Founder / product owner	1	Strategy, partners, fundraising, B2B sales
Full-stack engineers	2	Launch gaps, B2B API, AI features; on-call rota
Dispatch and support desk (24×7)	5	Three shifts plus cover. Watches the dispatch board, handles `NO_DRIVER_FOUND`, complaints and grievances
Partner acquisition and quality	1	Signs up and trains tow operators, audits documents, handles damage claims
Part-time / retained	—	Chartered accountant (GST, TDS, books), startup lawyer (documents, contracts), designer (as needed)

8. Milestones (summary of chapter 8)

Phase	Window	Exit criteria
0. Launch readiness	Weeks 0–6	Entity, GST, legal pages final, DLT SMS, production email, insurance, 30+ active partners, 24×7 desk staffed
1. Delhi pilot	Months 2–4	30+ jobs/day, median arrival < 35 min, completion > 90 %, grievance acknowledgement < 48 h
2. NCR + B2B	Months 4–9	Noida, Gurugram, Ghaziabad, Faridabad live; 2+ B2B contracts; Razorpay live; operator SaaS beta
3. Scale + AI wave 1	Months 9–18	200+ jobs/day; AI photo triage, voice/WhatsApp booking, smart dispatch; 2nd metro
4. Platform	Months 18–36	RSA membership, insurer/OEM APIs, 3+ cities, EBITDA-positive

9. Decisions needed from management

#	Decision	Recommendation	Why it matters
1	Legal entity	**Private limited company** before launch	The E-Commerce Rules expect a Companies Act company. It also gives limited liability and lets us raise investment (chapter 3)
2	Contracting model: who is the "seller" to the customer?	**Platform as principal** to start (matches the current code), revisited with the CA	Decides GST, TDS/TCS treatment, liability for service quality, and invoice design
3	Damage and liability model	Partners must hold insurance; platform investigates and facilitates; platform liability capped. Add platform goods-in-tow cover from year 2	The largest trust risk and the largest legal exposure
4	Named Grievance Officer	A named India-based person (founder at first)	Legal requirement; shown on the website
5	Brand name	Run a trademark search on "RoadSaathi" before spending on marketing	Avoids a forced rebrand later; changing the name is one setting in the product
6	Funding route	Lean self-funded pilot (≈ ₹50 L) **then** a seed round of ≈ ₹2.5–3 cr on pilot data	Pilot data (arrival times, repeat rate, CAC) makes fundraising far easier
7	B2C vs B2B emphasis	**B2C pilot to prove service quality, B2B to scale revenue**	Chapter 2: B2B and subscriptions are where the industry's profits are

10. Chapters in this pack

#	Chapter	File
2	Competitor analysis, India and global, with financials	`02-competitor-analysis.md`
3	Legal and regulatory map	`03-legal-regulatory-map.md`
4	Biggest obstacles and risk register	`04-risk-register.md`
5	AI features roadmap	`05-ai-roadmap.md`
6	Financial model and unit economics	`06-financial-model.md`
7	Go-to-market plan	`07-go-to-market.md`
8	Execution roadmap and launch checklist	`../product/ROADMAP.md`

Disclaimer: this plan is a business and research document, not legal, tax or investment advice. The legal and tax points in chapter 3 must be confirmed by an Indian startup lawyer and a chartered accountant before launch.

Chapter 2 — Competitor analysis (India and global)

Research date: November 2026. Public-company numbers come from company releases and filings as reported in the press; links are listed per row. Private-company numbers are **third-party estimates** and are marked as such. Currency is shown in the company's reporting currency to avoid conversion errors.

2.1 How the industry is structured

Roadside assistance worldwide is sold in three ways, and they have very different economics:

1. **Membership clubs** (AA and RAC in the UK, ADAC in Germany, AAA in the US). Customers pay an annual subscription. Revenue is recurring and prepaid, so the club can afford its own patrol fleet. **Highly profitable.**
2. **B2B2C assistance providers** (Agero, Allianz Partners, Europ Assistance, TVS Auto Assist, Urgently). Insurers and car makers pay per vehicle or per job. The provider runs a control room and **subcontracts the physical work to local tow operators**. Margins are thin, and the business depends on a few big contracts.
3. **Lead marketplaces and local operators** (Justdial, Google listings, thousands of independent towers). Highly fragmented. The information layer (Justdial) is very profitable; the towers themselves are small cash businesses.

RoadSaathi starts as a consumer marketplace (model 3, done properly), sells its supply network into model 2, and adds memberships (model 1) once the service is trusted.

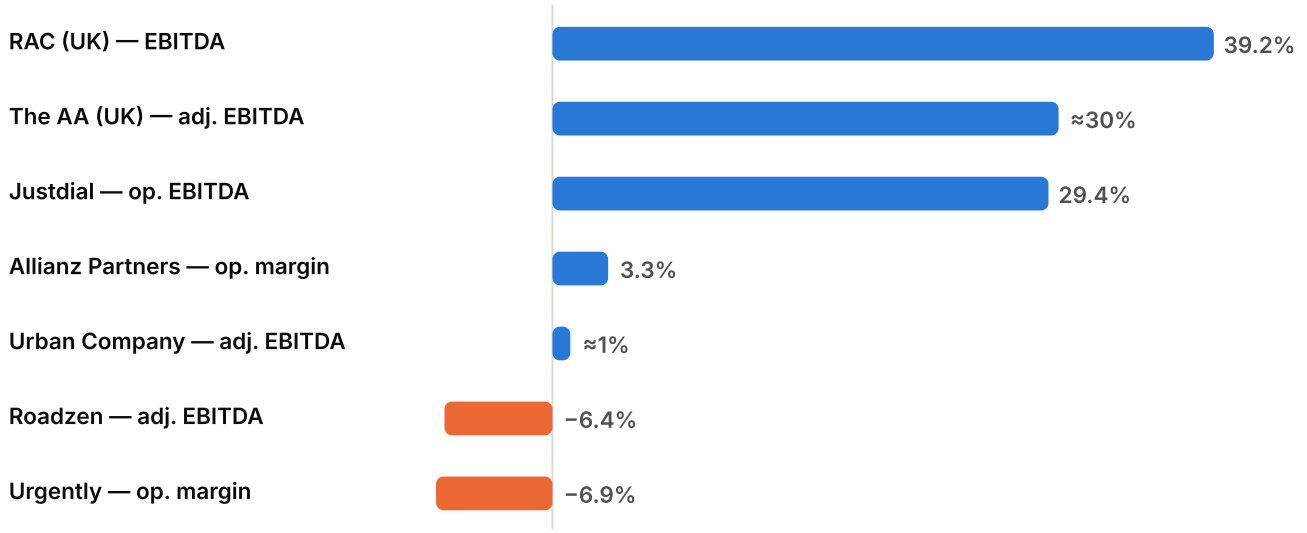
2.2 Global players

Company	Model	Latest reported revenue	EBITDA / profit	Why profit or loss
The AA (UK, private equity-owned)	Membership + insurance broking; own patrol fleet; 3.5 m breakdowns attended in 2025	**£1.5 bn** revenue (FY to Jan 2025, +12 %)	**Adj. EBITDA £450 m (≈ 30 %)**; operating profit £245 m	Recurring prepaid memberships, dense patrol network, cross-sell of insurance. Carries large private-equity debt. City AM, AA investors
RAC (UK, private equity-owned)	Membership; ~16 m members; breakdown, insurance, servicing	**£840 m** (FY2025, +7 %); H1 2026 £436 m	**EBITDA £329 m (39.2 % margin)**	Subscription pricing power and scale density. Breakdown is the core (£367 m of H1 2026 revenue). RAC FY2025 investor report, RAC H1 2026
Urgently (US, was Nasdaq: ULY)	Digital roadside platform for OEMs and insurers; asset-light, subcontracted providers	**US$129.2 m** (FY2025, −10 %)	**Net loss US$20.4 m**; operating loss US$8.9 m (improved from US$27.2 m). Reported margin ≈ 26 % in Q4 2025	Asset-light middleman with thin margins, contract losses and customer concentration, plus heavy debt. **Agreed to be acquired by Agero at US$5.50/share** (2026). Stockanalysis, SEC 8-K, Coverager
Agero (US, private)	B2B2C roadside and claims for insurers/OEMs; now owns Urgently	Not published; third-party estimates around **US$0.5–0.6 bn**	Not published	Scale player consolidating the US market. Buying Urgently shows that sub-scale digital players struggle alone. Lightsource profile (estimate)
HONK (US, private)	On-demand roadside marketplace + B2B partner programs	Estimates range **US$18–35 m** (unreliable)	Not published	~US$32–33 m raised in total. Shows consumer-plus-B2B marketplaces can survive but stay small without memberships. Tracxn
Allianz Partners (global assistance)	Travel insurance + assistance + roadside for OEMs/insurers; 36.9 m roadside cases in 2024	**€10.1 bn** total (2024, all lines)	**Operating profit €333.5 m (≈ 3.3 %)**	Huge volume but thin margins, because assistance is a low-margin service bundled for B2B clients. Allianz Partners 2024

2.3 Indian players

Company	Model	Revenue	Profit / EBITDA	Notes
Roadzen (Nasdaq: RDZN; large India operation)	Insurtech + roadside assistance + claims technology for insurers, OEMs and dealers	**US$55.0 m** (FY ending Mar 2026, +24 %); Q1 FY27 US$16.2 m (+49 %)	**Net loss US$22.5 m**; **adj. EBITDA loss US$3.5 m**; gross margin 61.3 %	Losses narrowing fast as one-off listing and integration costs fall away. The closest listed "Indian roadside tech" comparable. Roadzen FY2026 release, TradingView
ReadyAssist (Bengaluru, private)	RSA for OEMs/fleets/insurers + retail app	**₹12.8 cr+** (FY25, Inc42); Tracxn band ₹10–50 cr	Not reliably published	~US$6–7 m raised (pre-Series A of US$5 m). Our most direct Indian startup comparable. Inc42, Tracxn
TVS Auto Assist (Chennai; JV of TVS Automobile Solutions and the AA, UK)	B2B RSA for OEMs/insurers; 4,000+ authorised service providers, 1,795 locations, 28 states	Segment revenue not published	Not published	Shows the subcontracting model: the network *is* local operators like our partners. A potential **customer** for our Delhi supply, not only a competitor. Motor India
Allianz Partners India, **Europ Assistance India**	B2B RSA bundled with insurance and OEM warranties	Europ Assistance India: revenue band ₹1–100 cr (FY2023, Tofler)	Not published	Same model as above. Potential fulfilment clients. Tofler
Justdial (NSE: JUSTDIAL)	Local search and lead marketplace; where most Delhi towing customers search today	**₹1,141.9 cr** operating revenue (FY25, +9.5 %)	**Operating EBITDA ₹335 cr (29.4 %)**; net profit ₹584 cr, helped by ₹386 cr other income	Sells *leads*, not fulfilment: no trucks, no liability, very high margin. Our product replaces the "call five strangers" experience that follows a Justdial search. Business Standard
Urban Company (NSE-listed, 2025) — *model benchmark, not a competitor*	Managed marketplace for home services, standardised and trained partners	**₹1,144 cr** (FY25, +38 %)	Net profit ₹239.8 cr incl. ₹211 cr deferred tax credit; PBT ≈ ₹28.6 cr; adj. EBITDA ≈ ₹12 cr	Took about a decade to reach operating break-even. Proves Indian consumers pay for **standardised, trusted, priced-upfront** services. Medianama, Outlook Business
GoMechanic — *cautionary case*	Car-service aggregator	—	Collapsed January 2023: founders admitted "grave errors" in financial reporting, 70 % of staff laid off	Growth-at-all-costs plus weak financial controls. Lesson: audited books, a real-time ledger and governance from day one. Business Standard
Unorganised local towers (thousands in NCR)	Phone and cash; listed on Justdial and Google	Small cash businesses	—	Our **supply**, not just our competition. Most will earn more on the platform than off it (higher utilisation)

2.4 Margin comparison (latest reported year)



Blue = profit, orange = loss. Margins use each company's own reported measure (EBITDA, adjusted EBITDA or operating margin, as labelled), so they show the pattern rather than a strict like-for-like comparison. Sources are in the tables above.

2.5 What the numbers teach us

1. **Recurring revenue wins.** The two most profitable roadside businesses in the world (RAC 39 %, AA 30 %) earn annual memberships *before* any breakdown happens. → RoadSaathi adds an RSA membership in year 2 (chapter 7).
2. **Pure middlemen lose.** Urgently had a modern app and big OEM contracts but thin margins, customer concentration and debt. It shrank 10 % and was sold for about US$5.50 a share. → We must keep a real take rate, diversify clients, and avoid debt-funded growth.
3. **Information is cheap to sell, fulfilment is expensive.** Justdial earns 29 % EBITDA by selling leads; the towers do the hard work. → Our software (operator SaaS, white-label) should earn margin with no truck cost.
4. **Indian consumers pay for standardised trust.** Urban Company built a ₹1,144 cr business on upfront prices, trained partners and guarantees. It took about ten years, which shows how long patient capital takes. → Trip PIN, ratings, invoices and damage process are the product, not extras.
5. **Governance is existential.** GoMechanic's collapse came from accounting, not product. → Integer-paise ledger, immutable signed contracts, audit log (already built) plus audited books from year 1.
6. **B2B assistance companies are customers, not only rivals.** TVS Auto Assist, Allianz and Europ subcontract to local operators. A dense, tracked, verified Delhi network is something they would buy.

2.6 Our positioning statement

> For vehicle owners in Delhi-NCR who are stuck on the road, **RoadSaathi** is the towing and roadside service that shows the price before you book, tracks the truck live and only lets the verified driver start the job. Unlike calling numbers from a listing, every job comes with a trip PIN, a rating and a GST invoice. For insurers, dealers and fleets, RoadSaathi is a ready, tracked fulfilment network with an API.

Chapter 3 — Legal and regulatory map

Scope: what Indian law asks of a towing and roadside-assistance platform that (a) sells to consumers online, (b) works with independent tow operators and fleet companies, (c) handles personal and location data, and (d) collects money and issues GST invoices. Status is as of November 2026.

> **Not legal advice.** This is a research map to brief a Delhi startup lawyer and a chartered accountant, who must confirm every row before launch. Items marked ⚠ **confirm** are open questions where the answer depends on facts or on rules not yet notified.

3.1 Summary — what is mandatory before the first paying customer

#	Requirement	Law	Status in product	Owner
1	Incorporated entity (private limited recommended)	Companies Act 2013; Consumer Protection (E-Commerce) Rules 2020, Rule 4(1)(a)	Entity details are admin settings (placeholders today)	Founder + lawyer
2	Legal name, address, contact details and **named Grievance Officer** on the website; acknowledge complaints within 48 h, resolve within 1 month	E-Commerce Rules 2020, Rules 4 & 5; IT (Intermediary) Rules 2021	Grievance page exists; officer name field missing	Founder
3	Privacy Policy that matches real data practices	IT Act s.43A + SPDI Rules 2011; DPDP Act 2023 (phased)	Template exists; needs subprocessors, retention, cookie section	Lawyer + eng
4	Terms of Service, Cancellation & Refund policy shown **before** payment; no dark patterns	Consumer Protection Act 2019; E-Commerce Rules; Dark Patterns Guidelines 2023	Templates exist; acceptance is not recorded yet	Lawyer + eng
5	GST registration, tax invoices, returns	CGST/IGST Acts	Invoicing built (SAC, CGST/SGST/IGST split, FY numbering)	CA
6	DLT registration for SMS (sender ID + templates)	TRAI TCCCPR 2018	MSG91 adapter ready; DLT pending	Ops
7	Partner agreements (driver, fleet)	Indian Contract Act 1872; IT Act s.10A	Templates + e-sign flow built	Lawyer
8	Partner vehicle and driver compliance checks	Motor Vehicles Act 1988; CMVR 1989	Licence number field only; **document upload/verification missing**	Ops + eng
9	Security practices, 180-day logs, 6-hour incident reporting	CERT-In Directions (28 Apr 2022); SPDI Rules	Audit log exists; log-retention policy and incident plan to be written (see `docs/operations/INCIDENT-RESPONSE.md`)	Eng

3.2 Detailed map

A. Company, e-commerce and consumer protection

- **Consumer Protection (E-Commerce) Rules 2020.** These apply to entities selling goods *or services* over a digital network. Covered entities must be a Companies Act company (or a specified foreign entity). They must display legal name, address, customer-care and grievance-officer details, show the total price including taxes

before purchase, and handle complaints within the set timelines. The research you supplied notes amendments notified in September 2026 that take effect on **1 January 2027**, adding more disclosure and dark-pattern duties. ⚠ **confirm** the final amended text with counsel.

- **Guidelines for Prevention and Regulation of Dark Patterns, 2023.** No false urgency, drip pricing, forced sign-up, hidden subscription or confusing cancellation. *Product impact:* our night/surge surcharge is shown in the quote breakdown, which is good. The offer countdown is for drivers, not customers. Keep surge capped and visible.
- **Unfair contract terms.** The Consumer Protection Act 2019 lets consumer commissions strike down one-sided terms. A liability cap "limited to the fare" may not survive negligent damage to a car worth lakhs. Pair it with a real claims process and insurance (chapter 4, risk R5).

B. Information technology and data protection

- **IT Act 2000, s.43A and SPDI Rules 2011.** Covered businesses must publish a privacy policy and keep "reasonable security practices", documented in an information-security programme. ISO 27001 is one way to show this, but it is not mandatory. *Sensitive data we hold:* bank details and UPI IDs of partners, and passwords (none; we use OTPs).
- **IT (Intermediary Guidelines and Digital Media Ethics Code) Rules 2021.** Relevant to the extent the platform is an intermediary (user reviews, partner listings): publish rules and a user agreement, and name a grievance officer.
- **Digital Personal Data Protection Act 2023 and DPDP Rules 2025.** These come into force in phases. Under the commencement notification, the main notice, consent, security, breach-notification, children's-data and data-principal-rights duties apply from about **13 May 2027**. *Product impact (build before then):* itemised consent notice at booking and partner sign-up; consent records; withdrawal; **account deletion and data export** (missing today); retention schedule (e.g. booking data kept for the GST record period, analytics with no PII, as the tracking plan already requires); processor contracts with subprocessors (email, SMS, maps, Firebase, Razorpay, hosting); breach notification to the Data Protection Board and affected users.
- **Location data.** Drivers share continuous GPS while online. The partner agreement already covers this (clause 7.2). The privacy policy must say how long location history is kept. Today only the last point is stored on `Driver`; `BookingEvent` stores lat/lng at status changes.
- **CERT-In Directions 2022.** Report specified cyber incidents to CERT-In **within 6 hours** of noticing them. Keep ICT logs for a **rolling 180 days** (within Indian jurisdiction per the directions; see CERT-In FAQ on clarifications). Name a point of contact. *Product impact:* Vercel/Render logs are short-lived by default, so add a log drain to storage you control.
- **Electronic signatures. ⚠ important.** The current partner agreement says that typing a name plus an email OTP "constitutes a valid electronic signature under the IT Act 2000". Under the IT Act, a legally defined *electronic signature* means the methods notified in the Second Schedule (for example Aadhaar eSign or a digital signature certificate). A typed name with OTP is better described as an **electronically formed contract** (s.10A IT Act plus the Indian Contract Act), backed by our audit trail of time, IP and OTP. **Recommendation:** soften that sentence now, and offer Aadhaar eSign (via an ASP such as Leegality, Digio or Signzy) for fleet-company agreements.

C. Tax

- **GST registration.** Registration is compulsory above the threshold (₹20 lakh aggregate turnover for services in most states), and may be compulsory regardless of turnover for some e-commerce arrangements. ⚠ **confirm with CA.** Rate: the product defaults to **18 % under SAC 996799** (other supporting services to transport). The CA must confirm the SAC and rate per service (towing, on-spot repair and fuel delivery may differ).
- **Who supplies the customer?** This is the most important tax question:
  - *Platform as principal* (current code: platform issues the invoice on the full fare). The platform charges GST to the customer on the full value. Partners supply *to the platform*, and their own GST registration and reverse-charge questions follow. The platform deducts **TDS under s.194C** when paying partners above the thresholds.

    - *Platform as e-commerce operator (marketplace)*. Partners are the suppliers. The platform collects **GST TCS under s.52 CGST** (0.5 % CGST + 0.5 % SGST, or 1 % IGST) and deducts **income-tax TDS under s.194-O at 0.1 %** (rate from 1 Oct 2024). Towing is **not** currently a s.9(5) service, where the operator itself pays the GST.
    - The choice changes invoices, ledger and returns. The code supports the principal model today, and the ledger already separates gross, commission and net per job.
- **Cash jobs.** Partners collect the full fare including GST, and the platform recovers commission plus GST through settlements (already modelled in the ledger as "providers owe platform"). This needs a clear recovery process and credit limits (chapter 4, risk R4).

D. Transport and vehicles

- **Motor Vehicles Act 1988 and Central Motor Vehicles Rules 1989.** Tow and recovery trucks must be registered in the correct body type ("breakdown van / recovery vehicle" or "crane-mounted vehicle", per the CMVR vehicle-type notifications). They need valid fitness, permit (where applicable), insurance and PUC, and drivers need a licence of the right class. *Product impact:* the `Truck` table already has insurance, permit and fitness expiry fields. Add document upload, verification and expiry alerts before scale.
- **Motor Vehicle Aggregator Guidelines 2025 (MoRTH, 1 July 2025).** These are written for **passenger** transport aggregators: state licence, ₹5 lakh fee, driver training, AIS-140 trackers, and fare caps up to 2× base at peak. Towing is not passenger transport, so they probably do not apply directly. States adopt them through their own rules, though, and the surge-cap principle is a useful benchmark. ⚠ **confirm** with counsel and watch Delhi Transport Department notifications.
- **Accident vehicles and police.** Vehicles in accidents with an FIR or medico-legal case may need police release before removal. Operations must check before dispatching **accident recovery** (SOP in `docs/operations/OPERATIONS-SOP.md`).
- **Statutory towing.** Towing illegally parked vehicles is done by Delhi Traffic Police contractors under separate rules. It is a different market, and we must not imply any police authority.

E. Labour and gig workers

- **Code on Social Security 2020.** Came into force on **21 November 2025** together with the other labour codes. It defines gig workers, platform workers and aggregators. Aggregators will contribute **1–2 % of annual turnover, capped at 5 % of amounts paid to gig/platform workers**, to a social security fund. **The rate and start date are not yet notified.** Budget for 1 % of turnover from year 2 (chapter 6). Whether a tow-truck *business owner* (rather than a driver) counts as a "platform worker" is a ⚠ **confirm** item.
- **State gig-worker laws** (Rajasthan 2023, Karnataka 2025 and others) add registration and welfare-fee duties where we operate. Track this before expanding beyond Delhi.
- **Own employees.** EPF/ESI once headcount thresholds are crossed. An internal POSH committee once there are 10+ employees. Delhi Shops & Establishments registration for the office.

F. Payments

- **RBI Payment Aggregator directions.** Use a licensed PA (Razorpay). If the platform collects money on partners' behalf and pays them out, use the PA's split or settlement product (e.g. Razorpay Route) rather than pooling funds in our current account. ⚠ **confirm** under the final principal/marketplace model.
- **Refunds.** Commit to refund timelines in the Cancellation & Refund policy (5–7 working days today) and keep them.

G. Insurance

- Partners must carry vehicle insurance and should carry **goods-in-transit / goods-in-custody** cover for customer vehicles.
- The platform should buy **commercial general liability** and **cyber** cover once live, and consider platform-level goods-in-tow cover from year 2.

- **RSA memberships are not insurance** if sold as a prepaid service contract (call-outs included). Do not describe them as insurance or promise indemnity, which would bring them under IRDAI. ⚠ **confirm** the membership terms with counsel.

H. Intellectual property

- **Trademark search and filing** for the brand (Form TM-A, Class 39 transport/towing, Class 9/42 software). Fee: ₹4,500 per class for startups/small entities online.
- **Code ownership:** written IP assignment from every contributor (founders, employees, contractors), as required by Copyright Act s.19.
- **Open-source licences:** all dependencies are permissive (MIT/Apache/BSD/ISC). OpenStreetMap data and tiles need attribution under ODbL, which Leaflet shows by default. Public OSM tiles, Nominatim and OSRM are fair-use only, so move to a paid or self-hosted provider before real traffic.

3.3 Compliance calendar

When	Item
Before launch	Rows 1–9 of §3.1; insurance; trademark search; soften e-signature wording
Monthly / quarterly	GST returns (GSTR-1, GSTR-3B); TDS returns (quarterly); grievance register review
1 Jan 2027	E-Commerce Rules amendments take effect, so review the checkout, cancellation and pricing displays
By 13 May 2027	DPDP core obligations: consent manager readiness, deletion/export, breach process, processor contracts
On notification	Social-security fund contribution start date; any Delhi aggregator rules
Annually	ROC filings (AOC-4, MGT-7), income-tax return, statutory audit, policy review, security review

Chapter 4 — Biggest obstacles and risk register

Scoring: **Likelihood (L)** and **Impact (I)** on a 1–5 scale, **Score = L × I**. Review monthly; anything at 15 or above goes on the management agenda.

4.1 The five obstacles that decide success

1. **Supply density, 24×7.** A customer stuck at 2 a.m. in Dwarka needs a truck *near Dwarka at 2 a.m.* Arrival time is the product. With too few partners online, the dispatcher sees `NO_DRIVER_FOUND`, and one bad night can lose a customer for life. *Answer:* launch in a limited set of zones, guarantee minimum earnings for night partners in the pilot, and expand only when median arrival stays under 35 minutes.
2. **Customer acquisition cost vs. low frequency.** People need a tow rarely, perhaps once every few years. Paid search for "towing near me Delhi" is expensive, and our contribution is about ₹210 per job (chapter 6). *Answer:* SEO (locality pages are already built), partnerships with dealers, insurers, fleets, parking operators and housing societies, and memberships that turn a one-off into a recurring customer.
3. **Leakage (disintermediation).** After one job, the customer has the driver's number and may call directly next time. *Answer:* masked calling, platform-only benefits (invoice, damage cover, tracking, membership), a partner agreement clause against solicitation (already in clause 3.4), and making the platform the partner's best source of jobs so leaving costs them more than staying.
4. **Trust and damage claims.** One scratched luxury car or one rude driver can go viral. *Answer:* KYC'd partners, trip PIN (built), photo condition report at pickup and drop (to build, AI-assisted in chapter 5), mandatory partner insurance, a fast claims SOP, and ratings with automatic suspension.
5. **Incumbent B2B lock-in.** Most new cars and insured cars already come with "free" RSA from Allianz, Europ, TVS Auto Assist and others. *Answer:* sell to these companies as their Delhi fulfilment partner instead of fighting them, and win uncovered customers: older vehicles, two-wheelers, commercial vehicles, and people whose RSA failed them.

4.2 Risk register

ID	Risk	Category	L	I	Score	Mitigation	Owner	Early-warning signal
R1	Not enough partners online in a zone or hour; long arrivals	Operations	4	5	**20**	Zone-limited launch; night incentives; fleet-company partners; demand heatmaps (AI-4)	Ops lead	Time-to-assign > 5 min; `NO_DRIVER_FOUND` > 5 %
R2	CAC higher than contribution per job	Commercial	4	4	**16**	SEO, B2B channels, memberships, referral credits; stop any paid channel with CAC > 1.5× contribution	Founder	Blended CAC > ₹300
R3	Customers or partners transact off-platform	Commercial	4	4	**16**	Masked calls, platform-only value, anti-solicitation clause, quality bonuses	Founder	Repeat rate falling while partner job count steady
R4	Partners don't remit commission and GST on cash jobs	Financial	3	4	12	Credit limit per partner; offset against online earnings; auto-pause above a balance; push online payment	Finance	"Providers owe platform" balance > 14 days
R5	Customer vehicle damaged; claim exceeds partner's means	Legal/Trust	3	5	**15**	Partner insurance mandatory; photo evidence; claims SOP; platform cover from year 2	Ops lead	Any claim > ₹25,000

ID	Risk	Category	L	I	Score	Mitigation	Owner	Early-warning signal
R6	Safety incident (assault, theft, accident during tow)	Safety	2	5	10	KYC and police verification of drivers, trip PIN, live tracking, SOS contact, suspension policy	Ops lead	Any incident
R7	Regulatory change: aggregator licensing, social-security contribution, DPDP, e-commerce amendments	Legal	3	3	9	Compliance calendar (chapter 3); budget 1 % of turnover for social security; counsel on retainer	Founder	Gazette notifications
R8	Wrong GST/TDS/TCS model causes back taxes and penalties	Tax	3	4	12	Decide principal vs. marketplace with CA before launch; monthly reconciliations	Finance + CA	CA review findings
R9	Personal-data breach	Security	2	5	10	Security baseline (`docs/developer/SECURITY.md`), CERT-In incident plan, least-privilege admin roles, log retention	Eng lead	Unusual admin logins; rate-limit spikes
R10	OSM public endpoints throttle us (geocoding/routing)	Technology	4	3	12	Self-host Nominatim/OSRM, or switch to Google/Mappls (provider interface already exists)	Eng lead	429 errors from geo API
R11	SMS OTP delivery failures (DLT, operator filtering)	Technology	3	4	12	DLT registration; Firebase phone auth with email fallback (built); WhatsApp OTP later	Eng lead	OTP verify rate < 85 %
R12	Polling load grows with live tracking at scale	Technology	2	3	6	SSE/WebSockets + Redis for locations (scaling path in ARCHITECTURE.md)	Eng lead	DB CPU > 60 % sustained
R13	Fraud: fake bookings, GPS spoofing, collusive fake jobs to earn incentives	Fraud	3	3	9	OTP-verified accounts; PIN-gated start; GPS sanity checks; anomaly detection (AI-7)	Eng + Ops	Completed jobs with zero movement
R14	Key-person dependency (solo founder, single developer)	Organisation	4	4	**16**	Documentation (this pack), second engineer, shared credentials vault, runbooks	Founder	—
R15	Funding gap before break-even	Financial	3	5	**15**	Lean pilot first; 18–24 months runway at seed; monthly burn review	Founder	Runway < 9 months
R16	Brand/trademark conflict on "RoadSaathi"	Legal	2	3	6	Trademark search before marketing spend; brand is a single setting to change	Founder	Search result
R17	Accounting or governance failure (GoMechanic case)	Governance	1	5	5	Ledger is source of truth; monthly close; external audit; no booking of unrealised revenue	Finance	Ledger vs. bank mismatch
R18	24×7 desk burnout and attrition	People	3	3	9	Shift rota, automation of routine steps, clear escalation, fair pay	Ops lead	Missed SLAs at night

ID	Risk	Category	L	I	Score	Mitigation	Owner	Early-warning signal
R19	Weather surges (monsoon waterlogging, fog season) overwhelm supply	Operations	4	3	12	Seasonal partner standby agreements; surge cap ≤ 1.5× with clear display; pre-positioning	Ops lead	IMD alerts
R20	A big B2B client takes most of the volume (concentration, as at Urgently)	Commercial	2	4	8	No single client > 30 % of revenue; keep the consumer channel	Founder	Concentration report

4.3 Top risks heat view

Impact ↓ / Likelihood →	1	2	3	4	5
5	R17	R6, R9	R5, R15	R1	
4		R20	R4, R8, R11	R2, R3, R14	
3		R12, R16	R7, R13, R18	R10, R19	

Chapter 5 — AI features roadmap

Principle: every AI feature must move a business metric in the KPI table (chapter 7): arrival time, completion rate, CAC, damage disputes, support cost or leakage. AI that only looks impressive is out of scope. Every feature below says what data it needs. Most of that data is **already captured** by the platform: `BookingEvent` timelines, `DispatchOffer` accept/decline/expiry, driver GPS, reviews, ledger and analytics funnel.

5.1 Feature catalogue

#	Feature	What it does	Business value	Data needed (✓ = already captured)	Effort	Phase
AI-1	**Voice and WhatsApp booking agent (Hindi / English / Hinglish)**	A stranded customer calls or messages; an AI agent collects vehicle, problem and location (WhatsApp location pin), quotes the fare and creates the booking through the existing booking service, with a human takeover button	24×7 intake without extra night staff; many stranded users prefer to call; lower CAC from WhatsApp	Booking API ✓, rate cards ✓, FAQ ✓; WhatsApp Business API + speech-to-text/LLM provider	M	3
AI-2	**Photo triage and condition report**	Customer or driver uploads photos; a vision model identifies the vehicle class, visible damage, low ground clearance or EV badge, and recommends flatbed vs. wheel-lift. Pickup and drop photos are compared to flag new damage	Right truck first time; **evidence in damage disputes** (risk R5); insurer-ready claim pack	Photo upload (to build); vehicle catalog ✓; booking timeline ✓	M	3
AI-3	**Smart dispatch: ETA and acceptance prediction**	Predicts each driver's real arrival time (traffic, time of day, truck type) and probability of accepting, then offers the job to the driver set most likely to arrive fastest	Lower time-to-assign and arrival time (the core KPI); fewer dispatch rounds	`DispatchOffer` outcomes ✓, driver GPS ✓, `BookingEvent` timestamps ✓, OSRM/Google travel times ✓	M	3
AI-4	**Demand forecasting and driver pre-positioning**	Hour-by-zone forecast from history, weather (IMD rain alerts, fog), festivals, cricket matches and road closures; heatmap in the driver app and incentives for under-supplied zones	Fixes obstacle #1 (supply density); fewer `NO_DRIVER_FOUND`	Booking history ✓, areas ✓, weather API	M	3
AI-5	**Fair dynamic pricing with guardrails**	Suggests the surge multiplier per zone and hour from forecast vs. supply, capped (e.g. ≤ 1.5×) and always shown in the quote	Balances supply at peaks while staying compliant with dark-pattern rules and fare-cap norms	Pricing engine ✓ (surge is already a parameter), AI-4 forecast	S	4
AI-6	**Document AI for partner KYC**	OCR of driving licence, RC, insurance, permit and fitness certificates; checks against DigiLocker / Vahan / Sarathi where permitted; expiry reminders; face match of driver selfie to licence	Onboard partners in hours, not days; fewer expired-document risks (Motor Vehicles Act compliance)	Document upload (to build); `Truck` expiry fields ✓	M	2–3

#	Feature	What it does	Business value	Data needed (✓ = already captured)	Effort	Phase
AI-7	**Fraud and leakage detection**	Flags GPS spoofing (impossible speeds), completed jobs with no movement, repeated cancellations between the same customer and driver (collusion), and partners whose job count fell while customers reappear off-platform	Protects margins (risks R3, R13)	Driver GPS ✓, booking events ✓, ledger ✓, reviews ✓	M	4
AI-8	**Support and grievance copilot**	Summarises a booking's full timeline for the agent, drafts the reply in the customer's language, classifies the grievance and tracks the 48-hour legal acknowledgement deadline	Faster, consistent support; legal SLA compliance (chapter 3)	`BookingEvent` ✓, notification log ✓, audit log ✓	S	3
AI-9	**Insurance claim pack generator (B2B)**	One click produces a PDF and JSON pack (photos, timeline, GPS trail, invoice, recovery receipt) for the customer's insurer, available through the B2B API	Differentiator for insurer and OEM contracts; faster claims for customers	Invoice ✓, timeline ✓, photos (AI-2)	S	3
AI-10	**Predictive breakdown alerts (telematics partners)**	With OEM, fleet or OBD-device partners, predicts battery failure and other faults and offers a jump-start or service before the vehicle stops	Turns emergency jobs into scheduled ones; strong B2B story for fleets	Third-party telematics feed	L	5
AI-11	**Crash detection (opt-in)**	The PWA or a future app detects a likely crash from phone sensors and offers help and recovery, with emergency-contact sharing	Safety brand; earliest possible booking	Device sensors; explicit consent (DPDP)	L	5
AI-12	**Partner quality score and coaching**	Combines ratings, review text sentiment, cancellations, arrival accuracy and complaints into one score that drives job priority and coaching messages	Better service without manual policing	Reviews ✓, offers ✓, events ✓	S	3
AI-13	**Admin natural-language analytics**	Ops staff ask "cancellations in Dwarka last week by hour" and get a chart. The LLM writes read-only SQL against a safe view	Faster decisions without a data analyst	Analytics events ✓, transactional tables ✓	S	4
AI-14	**Bilingual content and SEO generation**	Drafts locality pages, FAQs and service pages in English and Hindi for human review	Organic acquisition (lower CAC)	Areas ✓, services ✓	S	2
AI-15	**Earnings forecast for partners**	Tells each partner expected earnings by zone and hour, and when to go online	Partner retention and supply at the right hours	Ledger ✓, AI-4	S	4

Effort: **S** ≈ 1–2 engineer-weeks, **M** ≈ 3–6, **L** ≈ 8+ including partnerships. Phases follow the execution roadmap: 2 = NCR + B2B (months 4–9), 3 = scale + AI wave 1 (9–18), 4 = platform (18–36), 5 = later.

5.2 Recommended first wave (months 9–18)

1. **AI-3 smart dispatch**, which directly improves the core KPI and uses data we already have.
2. **AI-2 photo condition report**, which builds trust, limits damage claims and sells to insurers (with **AI-9**).
3. **AI-1 WhatsApp and voice agent**, for night-time intake without hiring more desk staff.
4. **AI-8 support copilot**, a small effort that protects the 48-hour grievance SLA.

5.3 Architecture fit

- The modular monolith already isolates the places AI plugs into. Dispatch scoring goes inside `modules/dispatch` (`runDispatchRound` chooses drivers today by distance). Photo checks become a new `modules/media` module. The agent calls the same `createBooking` service the website uses, so business rules are never duplicated.
- LLM and vision calls go through one provider interface, the same pattern as `modules/geo/providers.ts`. That keeps the vendor swappable and makes cost, logging and redaction easy to control.
- **Human in the loop** for anything that affects money or a person's livelihood: price changes, partner suspension and claim decisions. AI recommends; staff approve. Each decision is written to the existing `AuditLog`.

5.4 AI governance (DPDP and trust)

- Collect only what each feature needs, with a purpose-specific consent line (e.g. "photos are used to record your vehicle's condition and resolve damage claims").
- Strip or mask personal data (names, phones, number plates where not needed) before sending to an external model. Record which vendors process data in the privacy policy's subprocessor list.
- Keep model inputs and outputs for audit for a defined period, then delete them.
- Publish a short "How we use AI" section in the privacy policy.

Chapter 6 — Financial model and unit economics

> **Every number in this chapter is a planning assumption, not a forecast or a promise.** The model exists to show *which levers matter* and to size the funding need. Replace assumptions with pilot data at month 3. Amounts are in **₹ lakh** (1 crore = 100 lakh) unless stated. GST is excluded from revenue (it is collected for the government).

6.1 Pricing already in the product

Rate cards are editable in Admin → Catalog → Rate cards. Seeded defaults:

Towing (base covers first 5 km)	Reference base	Per km after 5 km	Flatbed ×1.3 · Winch ×1.5 · Accident ×1.8
Two-wheeler	₹499	₹17–19	e.g. flatbed ₹650
Car	₹999	₹30–38	e.g. flatbed ₹1,300
SUV	₹1,299	₹38–48	e.g. flatbed ₹1,690
Commercial (LCV)	₹1,999	₹55–69	e.g. accident ₹3,600
Heavy (truck/bus)	₹3,999	₹99–113	e.g. accident ₹7,200

On-spot services are flat: jump-start ₹249–899, tyre ₹249–999, fuel delivery ₹299–599 (plus fuel), lockout ₹349–699, breakdown mechanic ₹349–1,199. Night (10 p.m.–6 a.m.) ×1.25; surge is off by default; 18 % GST is added.

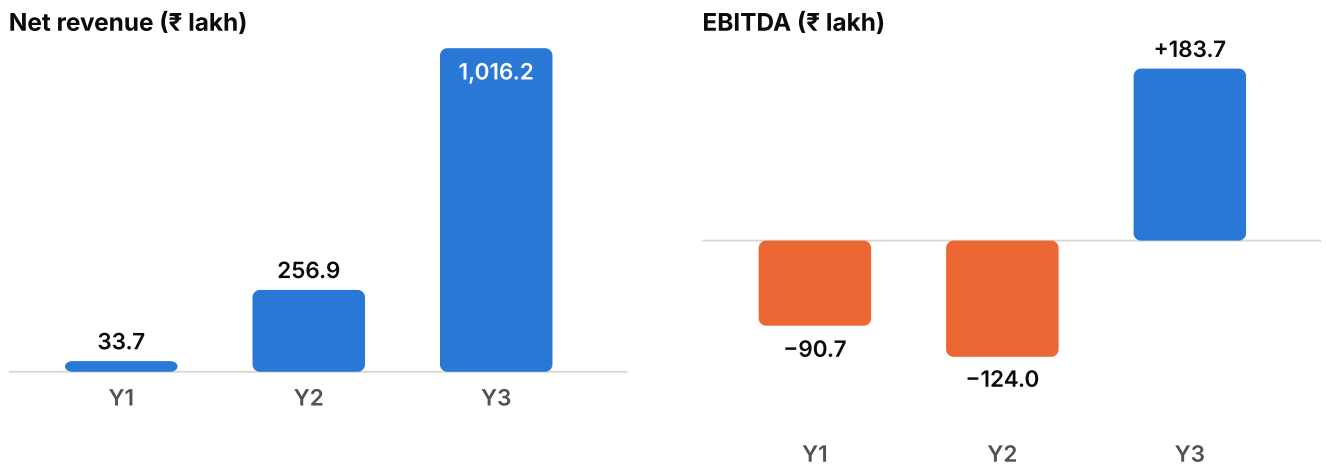
6.2 Unit economics per consumer job (assumptions)

Line	₹ per job	Assumption
Blended job value (taxable, before GST)	**1,400**	Mix of car/SUV towing (~~₹1,300–1,800) and on-spot services (~~₹350–600)
Platform commission (take rate)	**280**	20 % default (Admin → Pricing rules; overridable per partner)
Payment gateway	−10	30 % of jobs online × ~2 % of the GST-inclusive fare
SMS / OTP / maps	−6	DLT SMS ~₹0.2 each, OTPs, routing calls
Support and dispatch desk (variable share)	−40	Calls and chats per job
Damage-claim reserve	−14	1 % of job value
Contribution before marketing	**≈ 210**	75 % of commission
Customer acquisition cost (paid search, first job)	300–500	To be measured in the pilot

Reading: a customer bought with paid search only pays back after about two jobs. That is why the plan relies on **organic and partner channels, B2B volume and memberships** (chapter 7) rather than paid ads alone.

6.3 Three-year planning case

	Year 1 (Delhi pilot)	Year 2 (NCR + B2B)	Year 3 (scale, 2–3 cities)
Consumer jobs / day (average)	33 (ramp 5 → 80)	200	600
Consumer jobs / year	12,045	73,000	2,19,000
B2B fulfilment jobs / year	—	15,000	80,000
RSA members (₹1,199/yr)	—	2,000	20,000
Operator SaaS accounts	—	50 @ ₹1,000/mo	300 @ ₹1,200/mo
GMV, all jobs (taxable)	**₹1.7 cr**	**₹12.5 cr**	**₹43.3 cr**
Commission revenue	33.7	204.4	613.2
B2B margin (₹150/job)	—	22.5	120.0
SaaS revenue	—	6.0	43.2
Membership revenue	—	24.0	239.8
Net revenue	**33.7**	**256.9**	**1,016.2**
Variable costs (per-job costs, B2B handling, member call-outs paid to partners)	−8.4	−68.3	−297.3
Gross margin	25.3 (75 %)	188.6 (73 %)	718.9 (71 %)
People	−62	−180	−300
Marketing	−30	−80	−120
Tech infrastructure, SMS, maps	−6	−15	−35
Legal, CA, audit, insurance	−10	−15	−30
Office and other	−8	−20	−40
Social-security fund (assumed 1 % of revenue, from Y2)	—	−2.6	−10.2
EBITDA	**−90.7**	**−124.0**	**+183.7 (18 %)**
Cumulative cash need (before capex and working capital)	90.7	214.7	—



Planning case from the table above. Blue = positive, orange = negative. The two panels use different scales and should not be compared bar-to-bar.

6.4 Sensitivity (year-3 EBITDA, ₹ lakh)

Scenario	Net revenue	EBITDA
Planning case	1,016	**+184**
Take rate 15 % instead of 20 %	863	+32
Consumer volume 30 % lower (420/day)	832	+48
No membership product	776	+58
Average job value ₹1,200 instead of ₹1,400	929	+97
Half the B2B volume	956	+140
Upside: 750 jobs/day and 30,000 members	1,289	+360

Most sensitive levers: take rate, consumer volume and memberships. Any two adverse cases together push year 3 back to a loss, which is why the plan keeps fixed costs flexible (shift staff scale with volume) and reviews the model monthly.

6.5 Funding plan

Option	Amount	Covers	When
Lean pilot (self-funded / friends & family)	≈ ₹50 lakh	6 months of Delhi pilot at year-1 cost levels (≈ ₹58 L fixed less ≈ ₹10 L revenue)	Now
Seed round	≈ ₹2.5–3 cr	Years 1–2 cumulative need (≈ ₹2.15 cr) plus about 6 months' buffer	After 3 months of pilot data
Grants and schemes	Variable	Startup India Seed Fund Scheme, state startup policies (DPIIT recognition first)	In parallel

Use of seed funds (indicative): people 55 %, partner acquisition and marketing 25 %, technology and AI 10 %, compliance, insurance and buffer 10 %.

6.6 Assumptions to validate in the pilot (first 90 days)

1. Blended job value and service mix (towing vs. on-spot).
2. Partner acceptance of a 20 % commission. Benchmark: aggregators commonly take 15–30 %.
3. CAC by channel (search, SEO, dealers, societies, insurers).
4. Repeat and referral rate at 90 days.
5. Share of online vs. cash payments, and commission recovery days for cash jobs.
6. Willingness to pay for an annual membership (survey plus a ₹1 pre-booking test).

Chapter 7 — Go-to-market plan and KPIs

7.1 Customer segments (in launch order)

Segment	Why they need us	How we reach them	Phase
Delhi car and SUV owners without useful RSA (older cars, expired cover, RSA that failed them)	Need a fast, honest, priced tow now	Google search and SEO (locality pages already built), Google Business Profile, call-now CTAs	Pilot
Two-wheeler and three-wheeler owners (Delhi has more two-wheelers than cars; e-rickshaws are growing fast)	Cheap, quick on-spot help (puncture, battery, fuel)	Hindi-first pages, WhatsApp, petrol-pump and mechanic tie-ups	Pilot
Independent workshops, dealers, used-car dealers	Need reliable pickup and drop of customer cars	Partner sales; scheduled bookings (built)	Months 2–6
Fleets: cab operators, delivery fleets, logistics LCVs, rental cars	Downtime costs money; need invoices and tracking	B2B accounts, monthly GST invoicing, API	Months 4–9
Insurers, RSA companies, OEMs (Allianz, Europ, TVS Auto Assist, Roadzen, ReadyAssist and others)	Need a dependable tracked Delhi-NCR fulfilment network	Fulfilment contracts, API, insurer claim pack (AI-9)	Months 6–12
Housing societies, corporate parks, parking operators	Residents' and employees' vehicles break down on site	RWA partnerships, member pricing, QR posters	Months 3–9

7.2 Supply-side plan (partners), which comes first

Without supply there is no product. Target **30 active partners before launch** and **100 by month 4**, spread across the launch zones so every zone has at least 3 trucks online at night.

1. **List** every towing operator in launch zones from Justdial, Google Maps and transport-nagar yards.
2. **Visit**, demo the driver app (works on any Android phone, no install needed), and explain the earnings ledger.
3. **Offer a launch package:** 0 % commission for the first 20 jobs, a guaranteed night-shift minimum in the pilot, and priority dispatch for top-rated partners.
4. **Onboard** through the built flow: application → review → e-signed agreement → active. Add document upload (roadmap).
5. **Retain** with weekly settlements (built), fair dispatch, a partner WhatsApp group, a monthly top-partner award, and later the operator SaaS so partners run their whole business on RoadSaathi.

7.3 Demand channels and target CAC

Channel	Tactics	Target CAC
SEO	Area pages (`/areas/[slug]`), service pages, Hindi content, FAQ schema (JSON-LD already in code)	< ₹50
Google Business Profile	One profile per hub, reviews request after every job	< ₹50
Paid search	"towing near me", "car towing Delhi", "puncture repair near me" in launch zones only, at night	< ₹300
Partnerships	Dealers, workshops, RWAs, parking operators, petrol pumps, tyre shops, with referral fees	< ₹150

Channel	Tactics	Target CAC
B2B	Insurers, RSA companies, fleets	Sales cost per contract
Referral	₹100 credit for referrer and friend	< ₹200
PR and social	Launch story ("first priced-upfront towing in Delhi"), monsoon readiness campaigns	—

Marketing videos: the repo already contains a demo-film pipeline (`marketing/`) that produces product videos with voice-over from the real app.

7.4 Launch zones (Delhi pilot)

Start where breakdown density and partner supply overlap. Suggested first zones are South (Saket, Okhla), West (Dwarka, Janakpuri), Central (Connaught Place, Karol Bagh) and North (Rohini, Pitampura). Configure these in Admin → Service areas. NCR cities (Noida, Gurugram, Ghaziabad, Faridabad) are already seeded as **coming soon** with a notify-me form, and the `coming_soon_lead` event ranks which one to open next.

7.5 Pricing strategy

- Keep the **published rate card** as the core promise: no haggling, total shown before booking.
- Keep night surcharge (×1.25) and surge **capped and visible**. Plan a cap of 1.5× in line with dark-pattern rules and the aggregator-guideline benchmark of 2×.
- B2B: contracted per-job rates with monthly invoicing.
- Membership (year 2): ₹999–1,499/yr for N free call-outs plus discounted towing, benchmarked on insurer RSA add-ons.

7.6 KPIs (north star and guardrails)

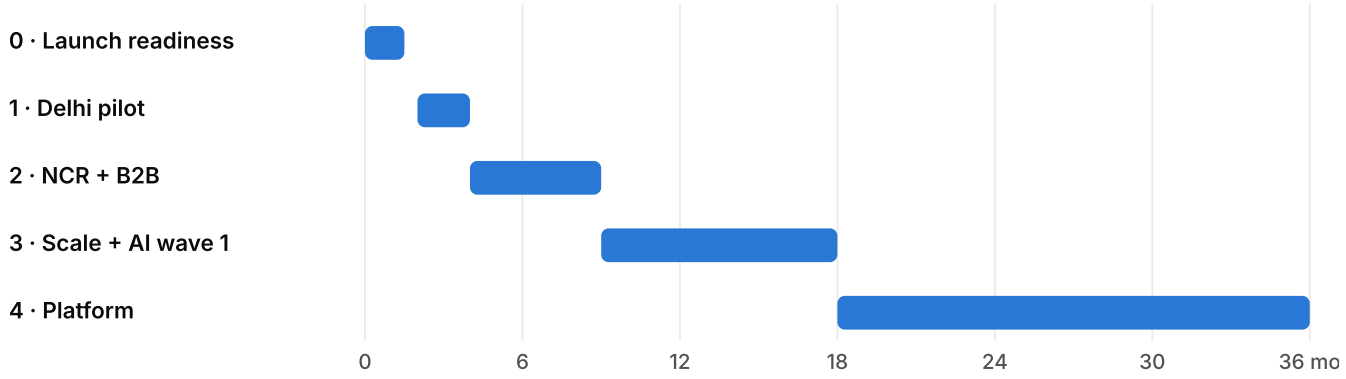
KPI	Definition	Pilot target	Where measured
North star: completed jobs per week	`Booking.status = COMPLETED`	Growth ≥ 10 % week on week in the pilot	Admin dashboard
Time to assign	`assignedAt − createdAt`	p50 < 3 min	Bookings / BookingEvent
Arrival time	ARRIVED event − ASSIGNED event	p50 < 35 min, p90 < 60 min	BookingEvent
Completion rate	Completed ÷ submitted	> 90 %	Analytics funnel
`NO_DRIVER_FOUND` rate	Share of bookings escalated	< 5 %	Dispatch board
Offer acceptance rate	Accepted ÷ offers sent	> 40 %	DispatchOffer
Cancellation rate (customer / partner)	Cancelled ÷ submitted	< 8 %	BookingEvent
Rating	Average review	≥ 4.5	Review
Booking funnel conversion	`booking_started → booking_submitted`	> 35 %	Analytics
CAC (blended)	Marketing spend ÷ new customers	< ₹300	Finance + analytics
90-day repeat or referral rate	Customers with 2+ jobs or a referral	> 15 %	Bookings
Grievance acknowledgement	Complaints acknowledged within 48 h	100 % (legal)	Support tool

KPI	Definition	Pilot target	Where measured
Cash commission recovery	Days to recover commission + GST on cash jobs	< 7 days	Ledger
Partner supply	Partner-hours online per zone at night	≥ 3 trucks per zone	Driver state

Chapter 8 — Execution roadmap and launch checklist

Owners: **F** founder · **E** engineering · **O** operations · **L** lawyer · **CA** chartered accountant.

8.1 Timeline



Months from go-ahead. Phases overlap slightly in practice; exit criteria, not dates, decide when the next phase starts.

8.2 Phase 0 — Launch readiness (weeks 0–6)

Business and legal

- [ ] Incorporate a private limited company; PAN, TAN, bank account (F, CA)
- [ ] GST registration; confirm SAC/rate per service and principal vs. marketplace model (CA)
- [ ] Trademark search and filing for the brand, classes 39 and 9/42 (L)
- [ ] Lawyer review of Terms, Privacy, Cancellation & Refund, Grievance, Cookie notice and both partner agreements (L)
- [ ] Appoint and publish the Grievance Officer (F)
- [ ] Insurance: commercial general liability and cyber; partner insurance requirements finalised (F)
- [ ] IP assignment agreements from all contributors (L)
- [ ] DPIIT Startup India recognition (optional, F)

Product and engineering (PRD v1.1 items L1–L11)

- [ ] Consent capture at booking and partner sign-up (E)
- [ ] Grievance officer settings fields and page (E)
- [ ] Partner KYC document upload and verification, with expiry blocking (E)
- [ ] Analytics consent banner (E)
- [ ] Contract template v2 with softened e-signature wording (L, E)
- [ ] Sentry, uptime monitor, log drain with 180-day retention (E)
- [ ] CI with type-check and tests (E)
- [ ] Production SMTP (SPF/DKIM), DLT SMS (MSG91) live, `EXPOSE_DEV_OTP=false`, rotate `AUTH_SECRET` and `CRON_SECRET` (E)
- [ ] Paid or self-hosted geocoding and routing before marketing spend (E)
- [ ] Load test: 50 concurrent bookings and 200 online drivers (E)
- [ ] Security review against `docs/developer/SECURITY.md` (E)

Operations

- [ ] Recruit and onboard 30+ partners across launch zones, with at least 3 night-capable trucks per zone (O)
- [ ] Staff the 24×7 desk (5 people, 3 shifts); train on `docs/operations/OPERATIONS-SOP.md` (O)

- [ ] Set rate cards, service areas, contact numbers and business details in Admin → Settings (O)
- [ ] Run 20 internal test bookings end to end, including night, cancellation and cash flows (O, E)

Exit criteria: every box above ticked; legal sign-off received; 30 active partners.

8.3 Phase 1 — Delhi pilot (months 2–4)

- Soft launch in 4 zones; Google Business Profile; SEO; small night-time paid search test.
- Weekly KPI review (chapter 7 table); fix the top three issues each week.
- Razorpay KYC and test mode; enable live mode once refunds (L10) ship.
- Collect the six pilot assumptions in chapter 6 §6.6.
- **Exit:** ≥ 30 jobs/day; p50 arrival < 35 min; completion > 90 %; rating ≥ 4.5; 100 % grievances acknowledged in 48 h.

8.4 Phase 2 — NCR and B2B (months 4–9)

- Open Noida, Gurugram, Ghaziabad and Faridabad in demand order (`coming_soon_lead`).
- B2B: accounts, API keys, status webhooks, monthly invoicing; sign 2+ contracts (dealers, fleets, an RSA company).
- Operator SaaS beta for 20 partners; push notifications; automated payouts; damage-claim workflow; photo condition report.
- Raise the seed round on pilot data.
- **Exit:** 150+ jobs/day; 2 B2B contracts live; CAC < ₹300 blended.

8.5 Phase 3 — Scale and AI wave 1 (months 9–18)

- AI-3 smart dispatch, AI-2 photo triage with AI-9 claim pack, AI-1 WhatsApp/voice agent, AI-8 support copilot.
- DPDP readiness by 13 May 2027: consent records, deletion/export, breach process, processor contracts.
- Second metro (choose by partner interest and demand data).
- Realtime (SSE/WebSockets) and Redis for locations; PostGIS for matching.
- **Exit:** 400+ jobs/day; contribution margin positive after marketing.

8.6 Phase 4 — Platform (months 18–36)

- RSA membership launch; insurer/OEM integrations; telematics pilots (AI-10).
- Three or more cities; per-city pricing and dispatch settings.
- **Exit:** EBITDA positive (chapter 6 planning case, year 3).

8.7 Engineering estimate for Phase 0 items

Item	Estimate (engineer-days)
Consent capture + records table	2
Grievance officer settings + page	0.5
KYC document upload (object storage, admin review, expiry blocking)	5
Analytics consent banner	1
Contract template v2	0.5
Monitoring, log drain, uptime	1.5

Item	Estimate (engineer-days)
CI pipeline	0.5
Production integrations (SMTP, DLT, secrets)	1.5
Geocoding/routing provider switch or self-host	2–4
Load test + fixes	3
Security review + fixes	2
Total	**≈ 20–22 engineer-days** (about 2 weeks for 2 engineers)